Inventory Management: Part 1

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INVENTORY

"The more inventory a company has, the less likely they will have what they need."

Taiichi Ohno (1912–1990)
Japanese Industrial Engineer

- Inventory is one of the most expensive assets of many companies. It represents as much as 50% of total invested capital.
- Operations managers around the globe have long recognized that good inventory management is crucial.
- On one hand, a firm can reduce costs by reducing inventory. On the other hand, production may stop and customers become dissatisfied when an item is out of stock.

INVENTORY

- All organizations have some type of inventory planning and control system.
- For example, a bank has methods to control its inventory of cash. A hospital has methods to control pharmaceuticals and blood supplies.
- In cases involving physical products, the organization must determine whether to produce goods or to purchase them.
- Once this decision has been made, the next step is to forecast demand.
- Then operations managers determine the inventory necessary to service that demand.

- **Basic inventory issues are**
  - **How much to order?**
  - **When to order?**

FUNCTIONS OF INVENTORY

- To provide a selection of goods for anticipated customer demand and to separate the firm from fluctuations in that demand.
- To decouple various parts of the production process. For example, if a firm's supplies fluctuate, extra inventory may be necessary to decouple the production process from suppliers.
- To take advantage of quantity discounts, because purchases in larger quantities may reduce the cost of goods or their delivery.
- To hedge against inflation and upward price changes.

Types of Inventory

- To accommodate the functions of inventory, firms maintain four types of inventories:
  - **Raw material inventory:** Materials that are usually purchased but are yet to enter the manufacturing process. This inventory can be used to separate suppliers from the production process.
  - **Work-in-process (WIP) inventory:** Products or components that are no longer raw materials but have yet to become finished products. WIP exists because of the time it takes for a product to be made (called cycle time). Reducing cycle time reduces inventory.

Types of Inventory

- **Maintenance/repair/operating supply (MRO) inventory:** MROs are inventories devoted to maintenance/repair/operating supplies necessary to keep machinery and processes productive. They exist because the need and timing for maintenance and repair of some equipment are unknown.
- **Finished-goods inventory:** An end item ready to be sold, but still an asset on the company's books. The final classification of inventory that is used for accounting purposes by manufacturing companies.

Inventory Models and Associated Costs

- Inventory control models assume that demand for an item is either independent of or dependent on the demand for other items.
- For example, the demand for a four-wheeler car is independent of the demand for two-wheeler motorcycle. However, the demand for helmets is dependent on the requirements of motorcycles.

Holding, Ordering, and Setup Costs

- **Holding costs** are the costs associated with holding or "carrying" inventory over time.
- Category of holding costs:
  - Housing costs (building rent or depreciation, operating cost, taxes, insurance) (3–10%)
  - Material-handling costs (equipment lease or depreciation, power, operating cost) (1–3.5%)
  - Labor cost (receiving, warehousing, security) (3–5%)
  - Investment costs (borrowing costs, taxes, and insurance on inventory) (6–24%)
  - Pilferage, scrap, and obsolescence (much higher in industries undergoing rapid change like tablets and smartphones) (2–5%)

Numbers indicate cost range as a percentage of inventory value

Ordering and Setup Costs

- **Ordering cost** includes costs of supplies, forms, order processing, purchasing, clerical support, and so forth. When orders are being manufactured, ordering costs also exist, but they are a part of what is called setup costs.
- **Setup cost** is the cost to prepare a machine or process for manufacturing an order. This includes time and labor to clean and change tools or holders. Operations managers can lower ordering costs by reducing setup costs and by using such efficient procedures as electronic ordering and payment.

Inventory Models for Independent Demand

- Inventory models that address two questions: (1) When to order and (2) How much to order?
- Different demand models are:
  - **Basic economic order quantity (EOQ) model**
  - **Production order quantity model**
  - **Quantity discount model**

Basic economic order quantity (EOQ) model

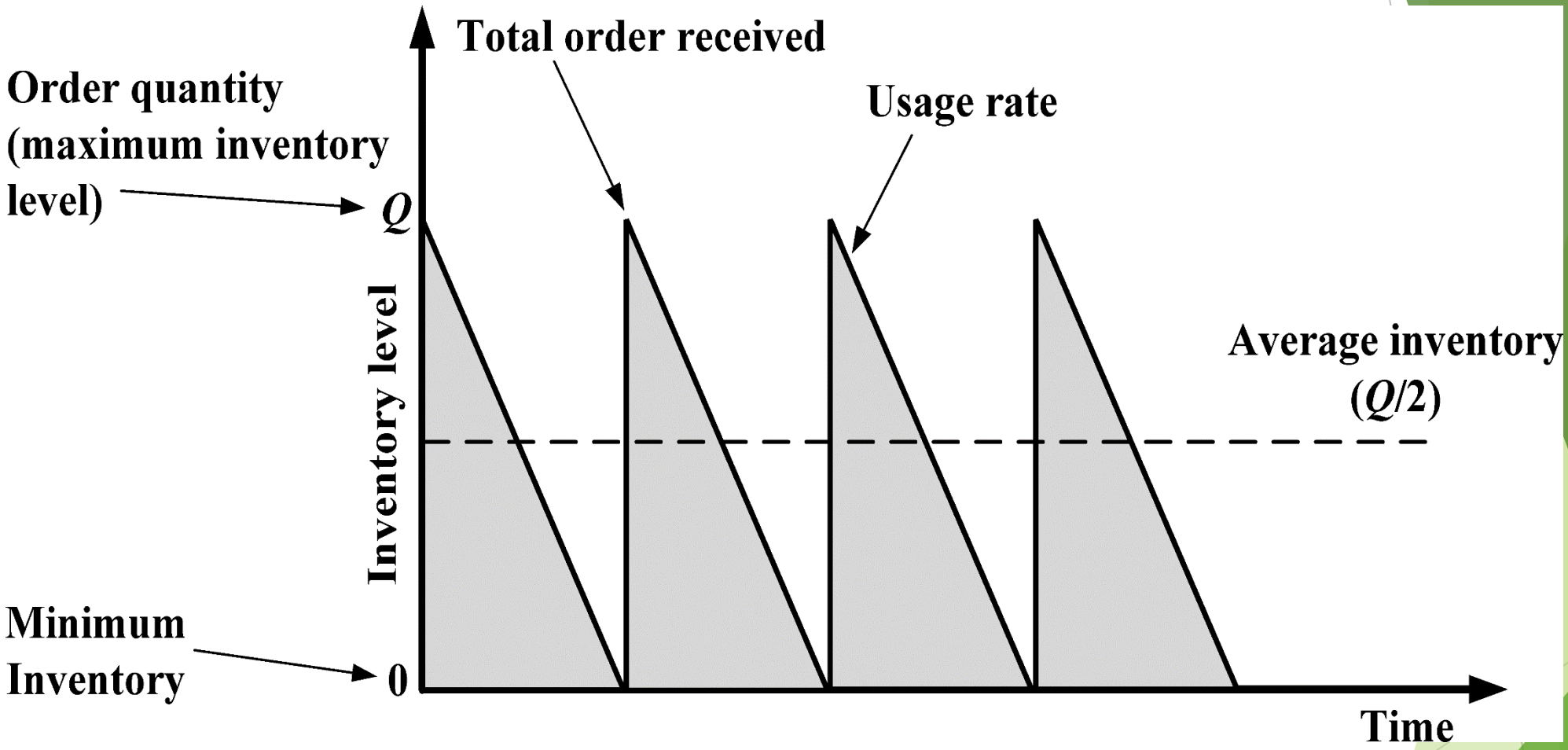
- EOQ model is one of the most commonly used inventory-control techniques. This technique is relatively easy to use. **It is based on several assumptions:**
  - Demand for an item is known, reasonably constant, and independent of decisions for other items.
  - **Lead time:** The time between placement and receipt of the order is known and consistent.
  - Receipt of inventory is instantaneous and complete. In other words, the inventory from an order arrives in one batch at one time.

Basic economic order quantity (EOQ) model

Assumptions

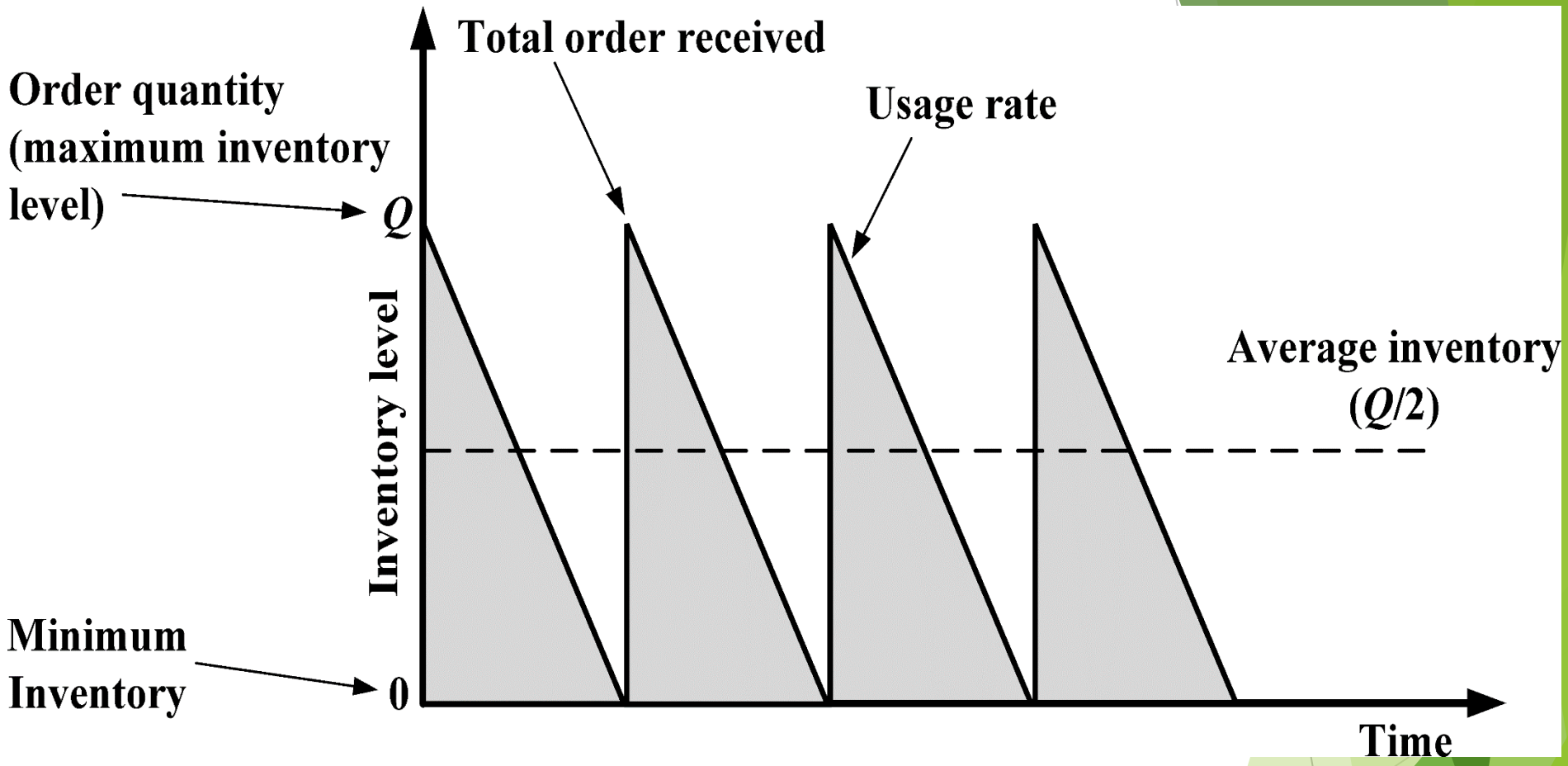
- Quantity discounts are not possible.
- The only variable costs are the cost of setting up or placing an order (setup or ordering cost) and the cost of holding or storing inventory over time (holding or carrying cost).
- Stockouts (shortages) can be completely avoided if orders are placed at the right time.

EOQ model



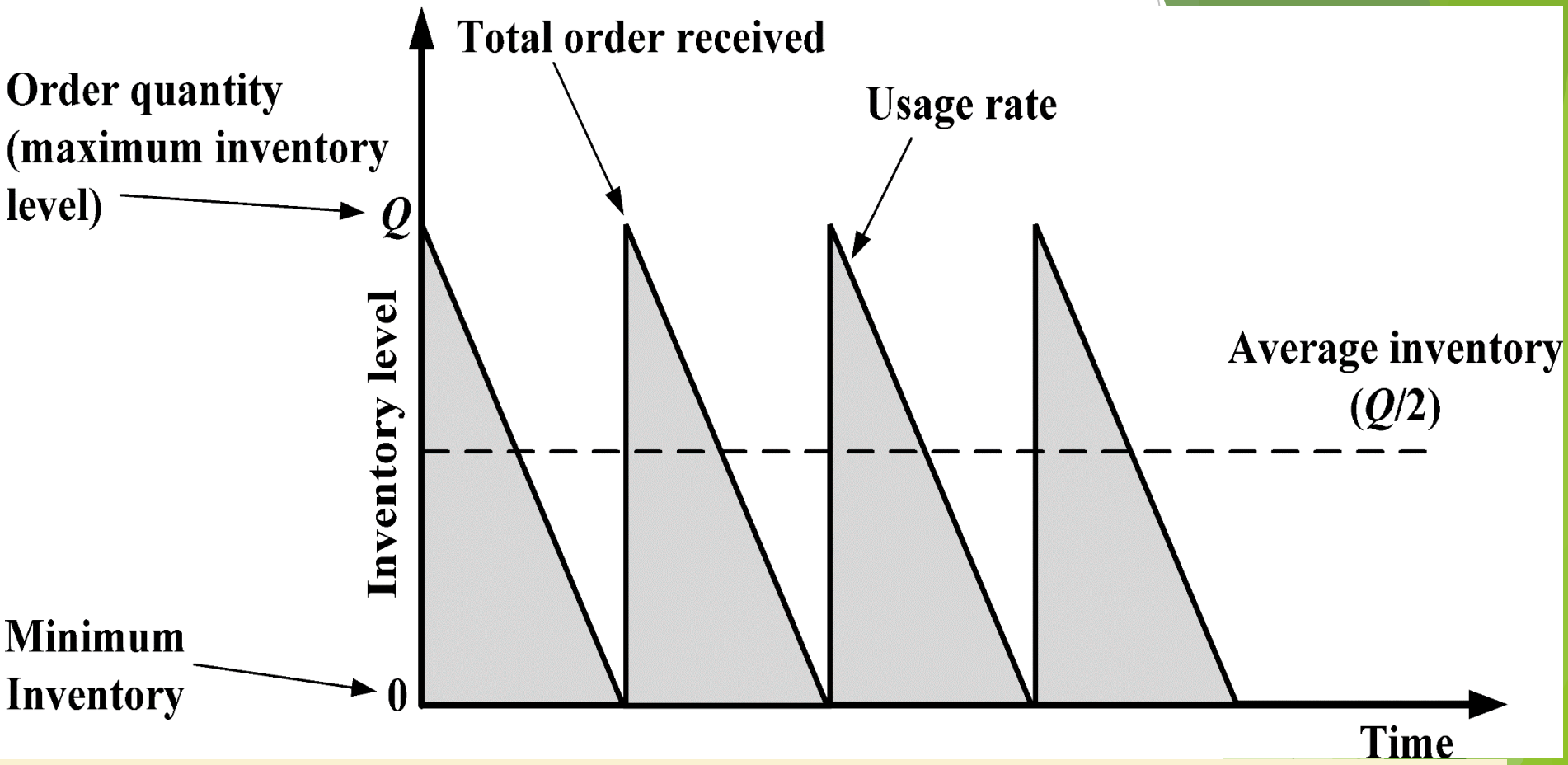
- The graph of inventory usage over time has a sawtooth shape as shown in Figure.
- *Q* represents the amount that is ordered.

EOQ model



- If the amount *Q* is 400 mobile phones, all 400 phones arrive at one time (when an order is received).
- Thus, the inventory level jumps from 0 to 400 phones. In general, an inventory level increases from 0 to *Q* units when an order arrives.

EOQ model



- As demand is constant over time, inventory drops at a uniform rate over time as shown by sloped lines.
- Each time the inventory is received, the inventory level again jumps to Q units (represented by the vertical lines). This process continues indefinitely over time

- The steps for developing EOQ model are as follows:
  1. Develop an expression for setup or ordering cost.
  2. Develop an expression for holding cost.
  3. Set setup (ordering) cost equal to holding cost.
  4. Solve the equation for the optimal order quantity.

- The variables to be used in the model are as follows:

  Q = Number of units per order

  Q^* = Optimum number of units per order (EOQ)

  D = Annual demand in units for the inventory item

  S = Setup or ordering cost for each order

  H = Holding or carrying cost per unit per year

- Annual setup/ordering cost = (Number of orders placed per year) × (Setup or order cost per order)

$$= \left(\frac{\text{Annual Demand}}{\text{Number of units in each order}} \right) (\text{Setup or order cost per order})$$

$$= \frac{D}{Q} S$$

- Annual holding cost = (Average inventory level) × (Holding cost per unit per year)

$$= \left(\frac{\text{Order quantity}}{2} \right) (\text{Holding cost per unit per year})$$

$$= \frac{Q}{2} H$$

- Optimal order quantity (Q^*) is found when annual setup (ordering) cost equals annual holding cost:

$$\frac{D}{Q}S = \frac{Q}{2}H$$

or $$2DS = Q^2H$$

or $$Q^2 = \frac{2DS}{H}$$

or $$Q^* = \sqrt{\frac{2DS}{H}}$$

Optimization method for finding EOQ

$$T_C = \frac{D}{Q}S + \frac{Q}{2}H;$$

$$\frac{\mathrm{d}T_C}{\mathrm{d}Q} = -\frac{D}{Q^2}S + \frac{H}{2} = 0 \Rightarrow Q^* = \sqrt{\frac{2DS}{H}};$$

$$\frac{\mathrm{d}^2T_C}{\mathrm{d}Q^2} = \frac{2D}{Q^3}S \text{ (a positive quanity; hence, minimum)}$$

- The expected number of orders placed during the year (N) and the expected time between orders (T) can be determined:

$$\text{Expected number of orders} = N = \frac{\text{Demand}}{\text{Order quantity}} = \frac{D}{Q^*}$$

$$\text{Expected time between orders} = T = \frac{\text{Number of working days per year}}{N}$$

Q: A company that markets face masks to pharmacies, would like to reduce its inventory cost by determining the optimal number of masks to obtain per order.

The annual demand is 5000 units; the setup or ordering cost is Rs. 1000 per order; the holding cost per unit per year is Rs. 50.

- **Solution:**

Optimal number of units per order:

$$Q^* = \sqrt{\frac{2DS}{H}}$$

$$= \sqrt{\frac{2 \times 5000 \times 1000}{50}}$$

$$= 448 \text{ units}$$

Q: In previous example, the company has a 250 working days. Find the number of orders (*N*) and the expected time between orders (*T*). The optimal order quantity was found out to be 448 units.

Solution:

$$\text{Expected number of orders} = N = \frac{\text{Demand}}{\text{Order quantity}}$$

$$= \frac{D}{Q^*} = \frac{5000}{448} = 11.16 \text{ orders per year}$$

A round-figure answer is 11 orders per year. However, let us consider two cases- 11 orders per year and 12 orders per year.

Case 1: If the number of orders is 11 orders per year

Annual set up cost = Rs 1000×11= Rs. 11,000

No. of quantities per order = 5000/11 = 454.55

Say, 455 units are set for 10 orders and 453 units are set for 11th order.

For this case, the holding cost is

$$\text{Holding cost} = \left(\frac{455}{2}\times 50\times\frac{10}{11}\right)+\left(\frac{453}{2}\times 50\times\frac{1}{11}\right)$$

$$= \text{Rs. } 11370.45$$

Total cost = Rs. (11,000+11,370.45)

= Rs. 22,370.45

Case 2: If the number of orders is 12 orders per year

Annual set up cost = Rs 1000×12= Rs. 12,000

No. of quantities per order = 5000/12 = 416.67

Say, 417 units are set for 11 orders and 413 units are set for 12th order.

For this case, the holding cost is

$$\text{Holding cost} = \left(\frac{417}{2}\times 50\times\frac{11}{12}\right)+\left(\frac{413}{2}\times 50\times\frac{1}{12}\right)$$

$$= \text{Rs. } 10416.67$$

Total cost = Rs. (12,000+10,416.67)

= Rs. 22,416.67

Final answer

- It is clear seen that sum of set up and holding cost is less for 11-order per year policy. Hence, we should procure the inventory in 11 orders. We will have to spend Rs. 22, 371/-.
- Expected time between orders=T=(number of working days per year)/ (number of orders placed in a year)= (250/11)=22.73≅23 working days

- The total annual variable inventory cost is the sum of setup and holding costs:

Total annual cost = Setup (ordering) cost + Holding cost

or $$T_C = \frac{D}{Q}S + \frac{Q}{2}H \qquad (1)$$

- Inventory costs may also be expressed to include the actual cost of the material purchased.
- If the annual demand (*D*) and the price per mask (*P*) are known, then Equation (1) becomes

$$T_C = \frac{D}{Q}S + \frac{Q}{2}H + PD$$

Reorder Points

- Based on the previous discussion, it can now be evaluated ***how much to order***. Now the second inventory question is ***when to order***.
- Simple inventory models assume that receipt of an order is instantaneous. It is assumed that a firm will place an order when the inventory level for that particular item reaches zero and that it will receive the ordered items immediately.

- In practice, the time between placement and receipt of an order, called lead time/delivery time, can be as short as a few hours or as long as months.
- The when-to-order decision is usually expressed in terms of a **reorder point** (ROP)

Reorder Points

- The reorder point (ROP) is given as

ROP = Demand per day (d) × Lead time for a new order in days (L)

This equation assumes that demand during lead time and lead time itself is constant.

When this is not the case, a safety stock (s_s) is added.

- The reorder point with safety stock becomes

ROP = Expected demand during lead time + Safety stock

- The demand per day (d) is given by

$$d = \frac{\text{Annual Demand}}{\text{Number of working days in a year}}$$

Production order quantity model

- In the previous inventory model, it is assumed that the entire inventory order was received at one time.
- Sometimes, the firm may receive its inventory over a period of time. Such cases require a different model.
- There can be two situations:
  1. When inventory continuously flows or builds up over a period of time after an order has been placed or
  2. When units are produced and sold simultaneously. Under these circumstances, we take into account daily production (or inventory-flow) rate and daily demand rate.
     - Under these circumstances, daily production (or inventory-flow) rate and daily demand rate are considered.

Production order quantity model

- The following symbols are used for production order quantity model:

  Q = Number of units per order
  H = Holding cost per unit per year
  p = Daily production rate
  d = Daily demand rate, or usage rate
  t = Length of the production run in days

$$\begin{pmatrix}\text{Maximum inventory}\\ \text{level}\end{pmatrix} = \begin{pmatrix}\text{Total production}\\ \text{during the production run}\end{pmatrix} - \begin{pmatrix}\text{Total used during}\\ \text{the production run}\end{pmatrix}$$

$$= pt - dt$$

However, $Q = pt$ and $t = \dfrac{Q}{p}$

Production order quantity model

$$\text{Maximum inventory level} = p\left(\frac{Q}{p}\right) - d\left(\frac{Q}{p}\right) = Q\left(1 - \frac{d}{p}\right)$$

$$\text{Annual inventory holding cost (or simply holding cost)} = \frac{\text{Maximum inventory level}}{2}(H) = \frac{Q}{2}\left(1 - \frac{d}{p}\right)H$$

- Optimal order (Q_p*) or production quantity when inventory is consumed as it is produced is found when annual setup (ordering) cost equals annual holding cost:

$$D\frac{S}{Q} = \frac{Q}{2}\left(1 - \frac{d}{p}\right)H$$

Production order quantity model

$$D\frac{S}{Q}=\frac{Q}{2}\left(1-\frac{d}{p}\right)H$$

or

$$Q^2=\frac{2DS}{H\left(1-d/p\right)}$$

or

$$Q_p^*=\sqrt{\frac{2DS}{H\left(1-d/p\right)}}$$

Production order quantity model

- ***Question*:** A company forecasts that its automobile parts will be 5,000 units next year, with an average daily demand of 4 units. However, the production process is the most efficient at 8 units per day. Hence, the company produces 8 per day but uses only 4 per day.

  The company wants to solve for the optimum number of units per order. The setup or ordering cost is Rs. 1000 per order; the holding cost per unit per year is Rs. 50.

- ***Solution:***

  Annual demand = D = 5,000 units

  Setup costs = S = Rs. 1000

  Holding cost = H = Rs. 50 per unit per year

  Daily production rate = p = 8 units daily

  Daily demand rate = d = 4 units daily

Production order quantity model

- ***Solution:***

Annual demand = D = 5,000 units

Setup costs = S = Rs. 1000

Holding cost = H = Rs. 50 per unit per year

Daily production rate = p = 8 units daily

Daily demand rate = d = 4 units daily

$$Q_p^* = \sqrt{\frac{2DS}{H\left(1-d/p\right)}} = \sqrt{\frac{2\times 5000\times 1000}{50\times\left(1-4/8\right)}} = 633\,\text{units}$$

The difference between the production order quantity model and the basic EOQ model is that the effective annual holding cost per unit is reduced in the production order quantity model because the entire order does not arrive at once.

Some items are made by metallic 3D Printer, for which the setup cost is `50603/- and it can make 100 components in one setup. The holding (or inventory carrying cost) per item is `1000/- and demand is 5 items per day. Starting from basic fundamentals (include derivation), show that economic order quantity (EOQ) is 22.495 items. Anupam rounded it off and wrote final answer as EOQ=22, but Tanya wrote EOQ=23. Who is correct? Is the setup cost equal to holding cost?

Given data: D=5×365 unit/year, S= ₹50603, H=₹1000×365 per (unit×year).
Hence,

$$Q = \sqrt{\frac{2 \times 5 \times 365 \times 50603}{1000 \times 365}} = \sqrt{\frac{10 \times 50603}{1000}} = \sqrt{506.03} = 22.495\,\text{unit}$$

The value 22.495 is bounded between 22 (y_1) and 23 (y_2). Assume that, y_1 will give lower total cost, i.e.,

$$\frac{DS}{y_1}+\frac{Hy_1}{2}<\frac{DS}{y_2}+\frac{Hy_2}{2}\Rightarrow DS\left(\frac{1}{y_1}-\frac{1}{y_2}\right)<\frac{H}{2}(y_2-y_1)\Rightarrow\frac{2DS}{H}<y_1y_2$$

This implies that for y_1 to give the lowest cost,

$$y_1y_2>(EOQ)^2.$$

For our case,

$$y_1y_2=22\times23=506;(\mathrm{EOQ})^2=(22.495)^2=506.025$$

Hence, the condition is not satisfied. Therefore, y_2 will give the lowest cost. Thus, Tanya is correct. However, the setup cost will not be equal to holding cost; holding cost will be slightly higher.